4.1

Welcome | {} miner_report.json U | selector_miner.py U

selector_miner.py > ...

```python
import xml.etree.ElementTree as ET
import glob
import json
import os
ANDROID_NS = '{http://schemas.android.com/apk/res/android}'
def mine_ids(path):
    report = []
    for file in glob.glob(path + "/**/*.xml", recursive=True):
        try:
            tree = ET.parse(file)
            root = tree.getroot()

            for elem in root.iter():
                res_id = elem.get(ANDROID_NS + 'id')
                if res_id:
                    report.append({
                        "file": os.path.basename(file),
                        "id": res_id.split('/')[-1],
                        "tag": elem.tag
                    })

        except ET.ParseError:
            continue
    return report
if __name__ == "__main__":
    data = mine_ids("../Artefakt02/decompiled_apk/res/layout")
    with open("miner_report.json", "w") as f:
        json.dump(data, f, indent=4)
    print(f"[OK] Extracted {len(data)} IDs")
```

```
PS C:\Users\tomas\Desktop\TM-99080\Artefakt04> python selector_miner.py
[OK] Extracted 664 IDs
```

4.2

```
PS C:\Users\tomas\Desktop\TM-99080\Artefakt04> python .\stability_auditor.py
Audit Complete. Verdict: LOW_RISK
```

stability_report.json > { } metrics


```
{
    "metrics": {
        "LinearLayout": "18.72% (318 objects)",
        "TextView": "31.96% (543 objects)",
        "androidx.appcompat.view.menu.ActionMenuItemView": "0.06% (1 objects)",
        "androidx.appcompat.widget.ActionMenuView": "0.06% (1 objects)",
        "androidx.appcompat.widget.ActionBarContextView": "0.12% (2 objects)",
        "ImageView": "4.0% (68 objects)",
        "view": "0.53% (9 objects)",
        "FrameLayout": "2.35% (40 objects)",
        "ScrollView": "1.71% (29 objects)",
        "androidx.appcompat.widget.ButtonBarLayout": "0.06% (1 objects)",
        "Button": "16.83% (286 objects)",
        "android.widget.Space": "0.24% (4 objects)",
        "androidx.appcompat.widget.AlertDialogLayout": "0.06% (1 objects)",
        "include": "0.71% (12 objects)",
        "View": "0.88% (15 objects)",
        "androidx.core.widget.NestedScrollView": "0.06% (1 objects)",
        "androidx.appcompat.widget.DialogTitle": "0.06% (1 objects)",
        "androidx.appcompat.view.menu.ListMenuItemView": "0.18% (3 objects)",
        "Space": "0.06% (1 objects)",
        "androidx.appcompat.widget.FitWindowsLinearLayout": "0.12% (2 objects)",
        "androidx.appcompat.view.menu.ExpandedMenuView": "0.06% (1 objects)",
        "CheckBox": "1.88% (32 objects)",
        "RelativeLayout": "0.77% (13 objects)",
        "RadioButton": "0.41% (7 objects)",
        "merge": "0.29% (5 objects)",
        "androidx.appcompat.widget.ContentFrameLayout": "0.06% (1 objects)",
        "androidx.appcompat.widget.ViewStubCompat": "0.12% (2 objects)",
        "androidx.appcompat.widget.FitWindowsFrameLayout": "0.06% (1 objects)",
        "androidx.appcompat.widget.ActionBarOverlayLayout": "0.06% (1 objects)",
        "androidx.appcompat.widget.ActionBarContainer": "0.06% (1 objects)",
        "androidx.appcompat.widget.Toolbar": "0.06% (1 objects)",
        "ImageButton": "0.47% (8 objects)",
        "requestFocus": "2.3% (39 objects)",
        "EditText": "2.3% (39 objects)",
        "ListView": "0.94% (16 objects)",
        "ViewFlipper": "0.06% (1 objects)",
        "Spinner": "0.59% (10 objects)",
        "SeekBar": "0.65% (11 objects)",
        "AutoCompleteTextView": "0.35% (6 objects)",
        "MultiAutoCompleteTextView": "0.06% (1 objects)",
        "ToggleButton": "0.18% (3 objects)",
        "Chronometer": "0.12% (2 objects)",
        "RadioGroup": "0.12% (2 objects)",
        "NumberPicker": "0.06% (1 objects)",
        "io.appium.android.apis.view.LabelView": "0.18% (3 objects)",
        "TimePicker": "0.06% (1 objects)",
        "io.appium.android.apis.view.DraggableDot": "0.24% (4 objects)",
        "WebView": "0.18% (3 objects)",
        "fragment": "0.24% (4 objects)",
        "ProgressBar": "0.35% (6 objects)",
```


Analiza statyczna wykazała, że Indeks Dominacji Klas (CDI) dla komponentów w analizowanej aktywności nie przekracza progu krytycznego 50%. Oznacza to, że struktura interfejsu użytkownika jest relatywnie zróżnicowana pod względem typów komponentów, co zmniejsza ryzyko nadmiernej redundancji elementów tej samej klasy. W rezultacie stosowanie selektorów opartych na typie komponentu, takich jak By.className, jest mniej problematyczne niż w przypadku wysokiej dominacji jednej klasy.

4.3

```
PS C:\Users\tomas\Desktop\TM-99080\Artefakt04> python .\selector_game.py
=== INTERAKTYWNY KREATOR SELEKTORÓW ===
1. Podaj wartość 'id' z raportu (np. lunch): toggle_use_logo
2. Podaj wartość 'tag' z raportu (np. RadioButton): Button

Wynik wyszukiwania: Znaleziono 1 dopasowań.
>>> STATUS: ZALICZONE! Twój selektor jest unikalny. <<<
```

4.4

```
PS C:\Users\tomas\Desktop\TM-99080\Artefakt04> python .\a11y_check.py
Znaleziono 1 luk w dostępności. Raport zapisany.
```

{} a11y_report.json > ...


```
[
    {
        "file": "layout\\text_fields.xml",
        "id": "edit1Text",
        "text": "@string/edit_text_password",
        "issue": "Brak atrybutu contentDescription"
    }
]
```


4.5

```
PS C:\Users\tomas\Desktop\TM-99080\Artefakt04> python .\mock_driver_test.py
=== URUCHAMIANIE TESTU INTEGRACYJNEGO (PYTHON MOCK DRIVER) ===
←[92m[PASS] Selektor zweryfikowany pozytywnie.←[0m
[INFO] Mock Driver: Nawiązywanie połączenia z sesją...
[INFO] Mock Driver: Element znaleziony w czasie 12ms.
[INFO] Mock Driver: Akcja 'click' wykonana pomyślnie.

========================================
>>> WYNIK KOŃCOWY BLOKU 4: PASS <<<
========================================
```

test_execution.log

```
FINAL TEST RESULT: PASSED
TIMESTAMP: 2026-03-10 20:03:55
VALIDATED DATA:
PROJEKT SELEKTORA:
ID: toggle_use_logo
TAG: Button
STATUS: ZALICZONE
```